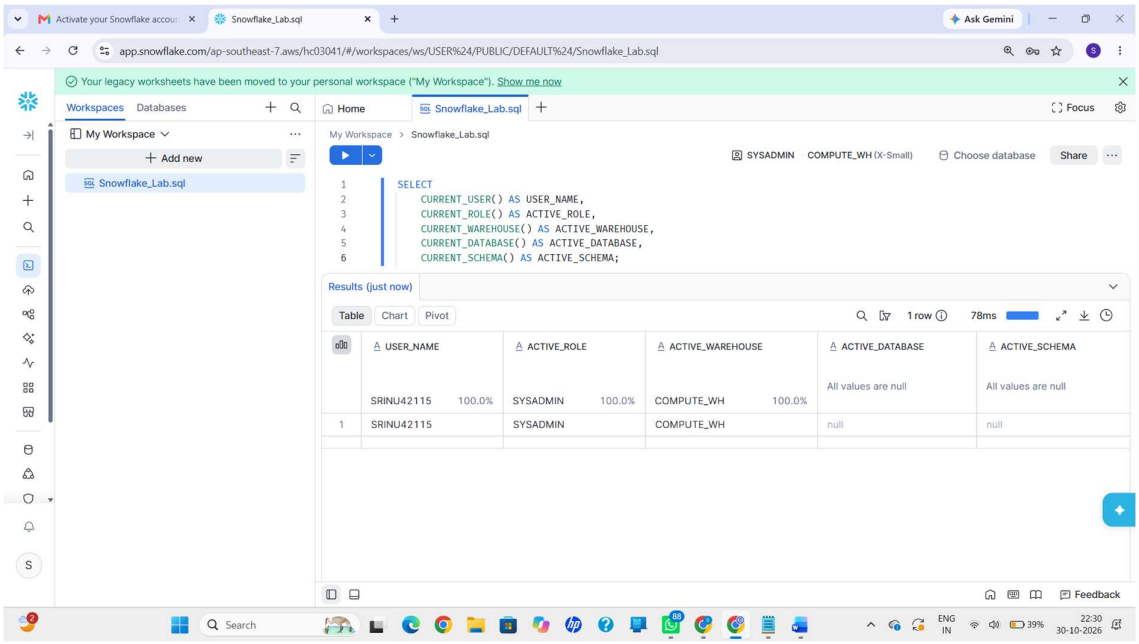
SNOWFLAKE SQL

PART 1: DATABASE AND SESSION MANAGEMENT

Step 1: Check Current Session Information

```
SELECT
    CURRENT_USER() AS USER_NAME,
    CURRENT_ROLE() AS ACTIVE_ROLE,
    CURRENT_WAREHOUSE() AS ACTIVE_WAREHOUSE,
    CURRENT_DATABASE() AS ACTIVE_DATABASE,
    CURRENT_SCHEMA() AS ACTIVE_SCHEMA;
```



Explanation:

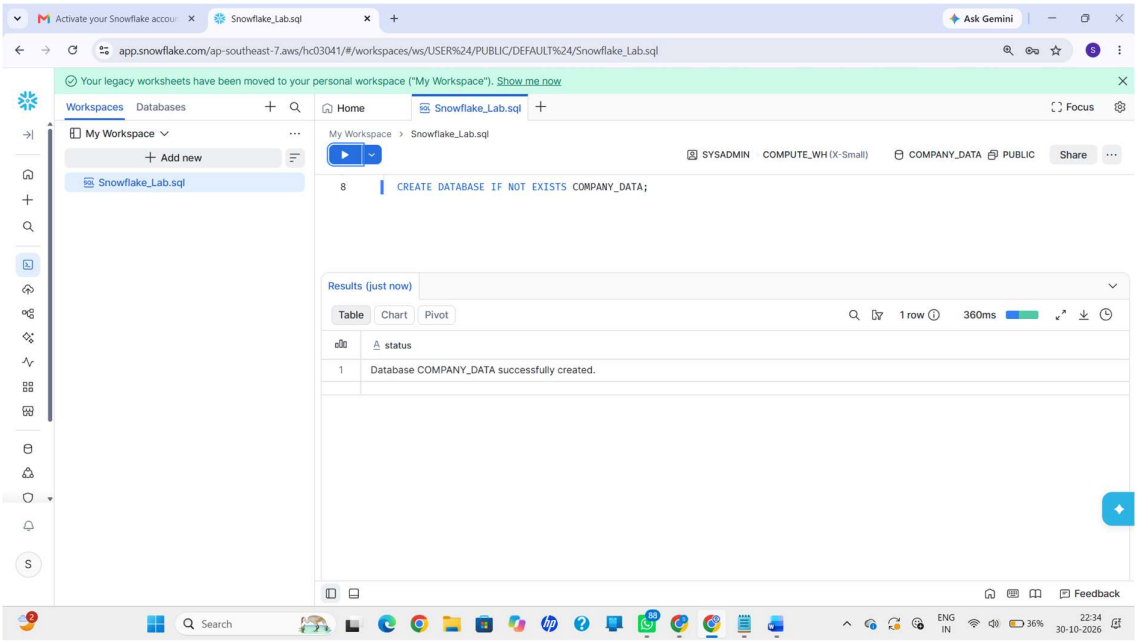
- CURRENT_USER() – Displays the logged-in user.
- CURRENT_ROLE() – Displays the active role.
- CURRENT_WAREHOUSE() – Displays the active warehouse.
- CURRENT_DATABASE() – Displays the selected database.
- CURRENT_SCHEMA() – Displays the selected schema.

Step 2: Create a Database

```
CREATE DATABASE IF NOT EXISTS COMPANY_DATA;
```

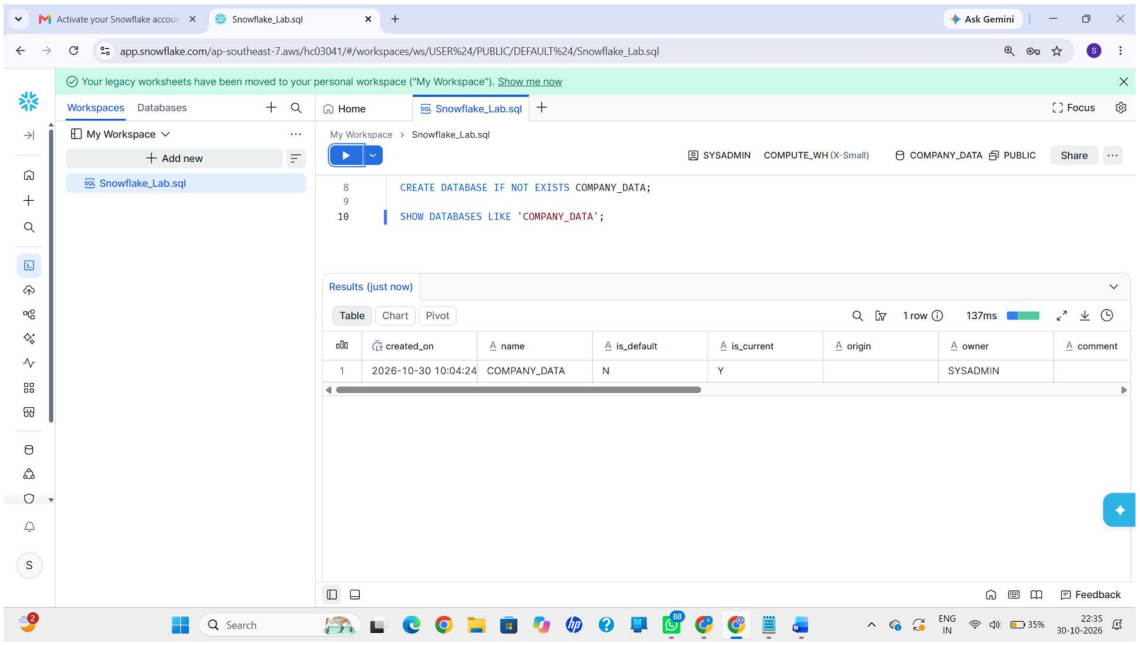
Explanation:

- **CREATE DATABASE creates a new database.**
- **IF NOT EXISTS prevents an error if the database already exists.**
- **COMPANY_DATA is the database name.**



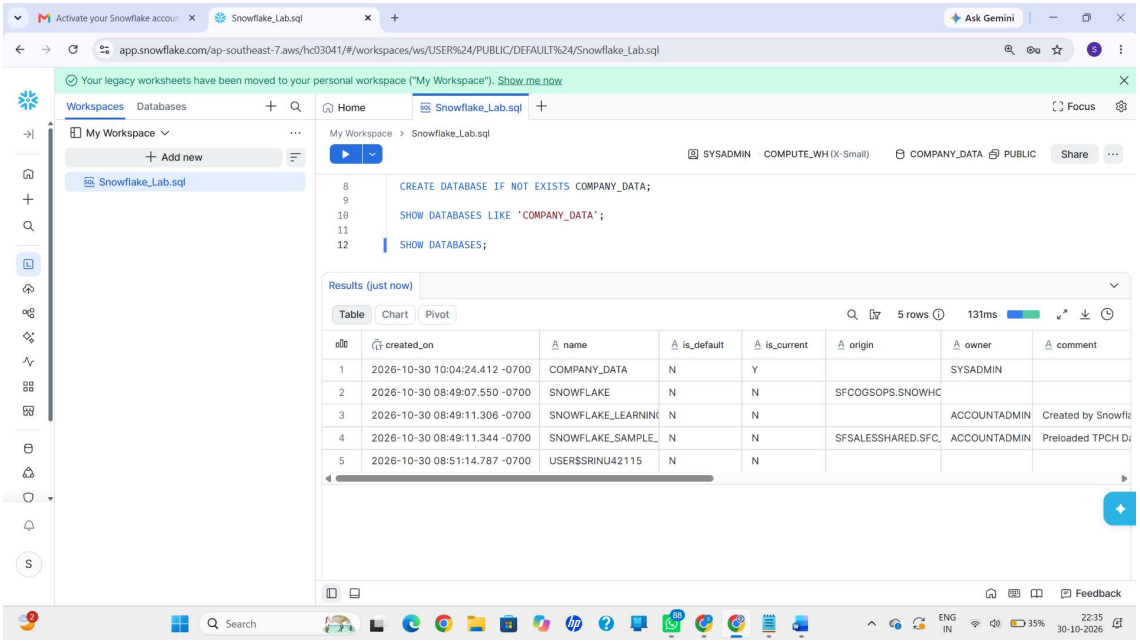
Verify the database:

```
SHOW DATABASES LIKE 'COMPANY_DATA';
```



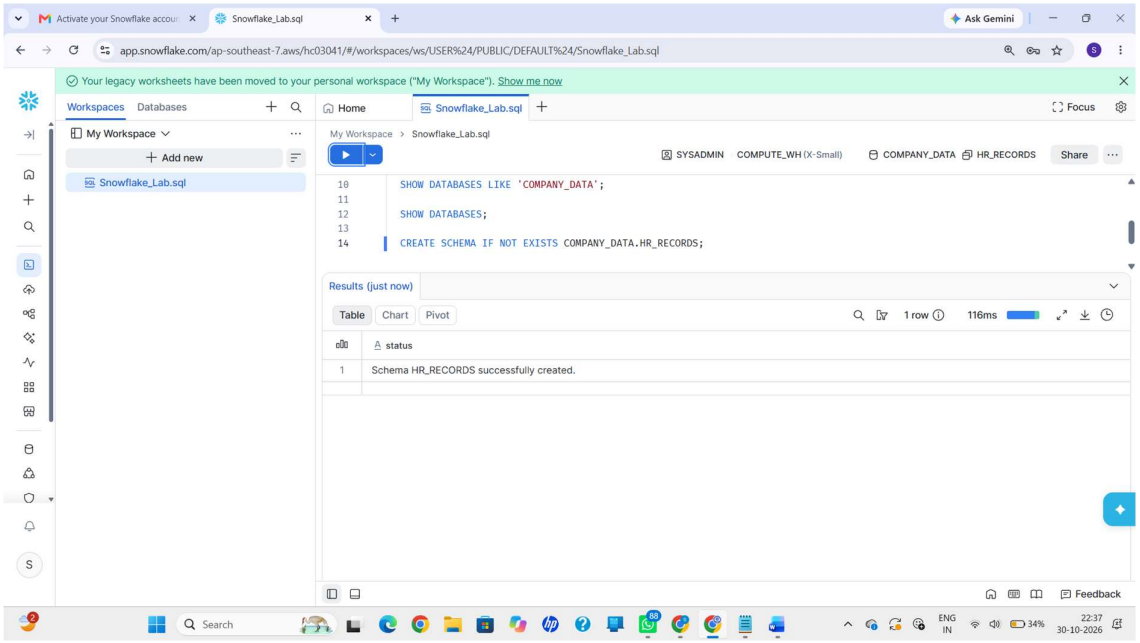
Display all databases:

```
SHOW DATABASES;
```



Step 3: Create a Schema

```
CREATE SCHEMA IF NOT EXISTS COMPANY_DATA.HR_RECORDS;
```



Explanation:

A schema is a logical container inside a database. It organizes tables, views, stages and other database objects.

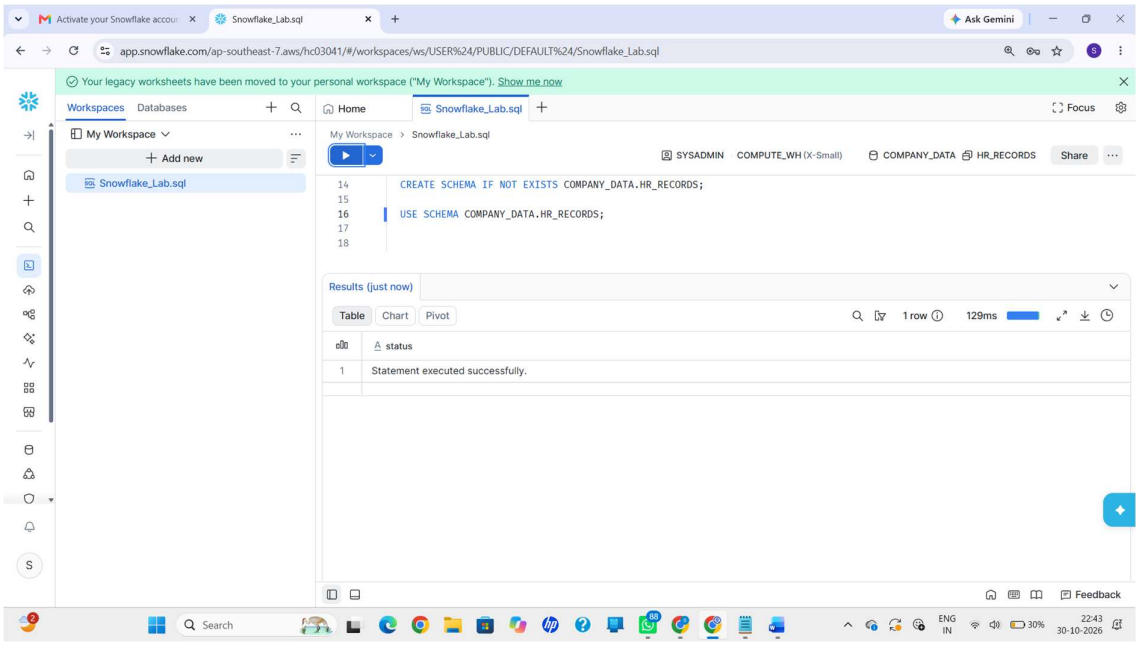
Here:

- Database: COMPANY_DATA
- Schema: HR_RECORDS

Select the schema:

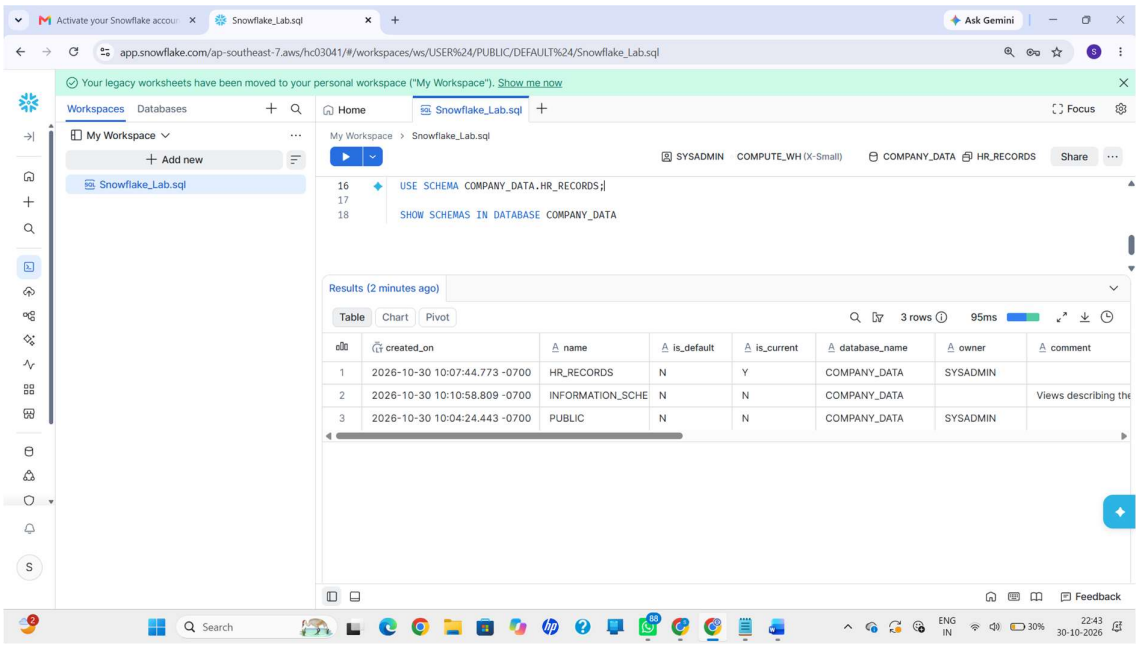
```
USE SCHEMA COMPANY_DATA.HR_RECORDS;
```

This tells Snowflake that you want to work inside **HR_RECORDS**.



Verify:

```
SHOW SCHEMAS IN DATABASE COMPANY_DATA;
```



PART 2: VIRTUAL WAREHOUSE MANAGEMENT

Step 4: Create a Virtual Warehouse

```
CREATE WAREHOUSE IF NOT EXISTS STAFF_WH

WAREHOUSE_SIZE = 'XSMALL'

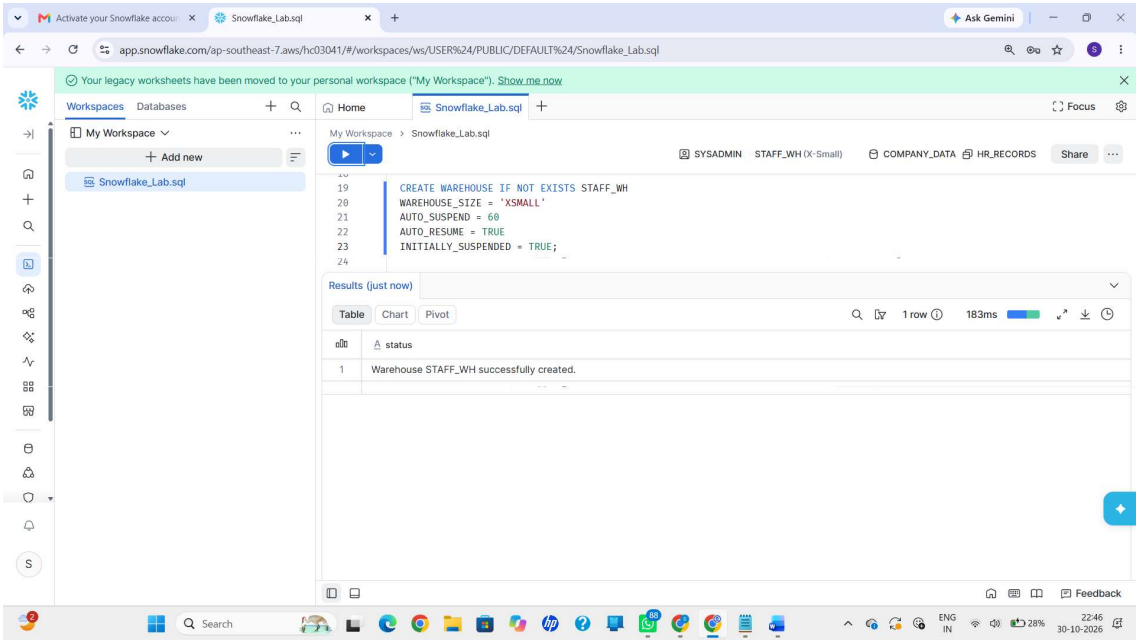
AUTO_SUSPEND = 60

AUTO_RESUME = TRUE

INITIALLY_SUSPENDED = TRUE;
```

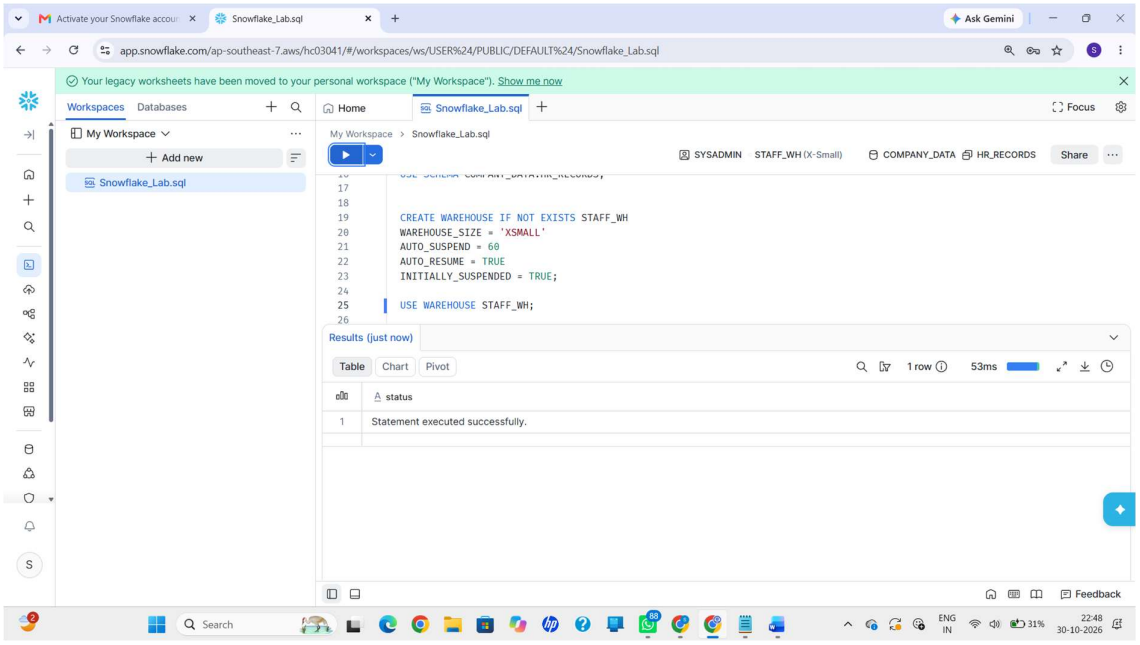
Explanation:

- WAREHOUSE_SIZE – Specifies the computing capacity.
- XSMALL – Smallest standard warehouse size.
- AUTO_SUSPEND = 60 – Suspends after 60 seconds of inactivity.
- AUTO_RESUME = TRUE – Automatically resumes when needed.
- INITIALLY_SUSPENDED = TRUE – Creates the warehouse in suspended mode.



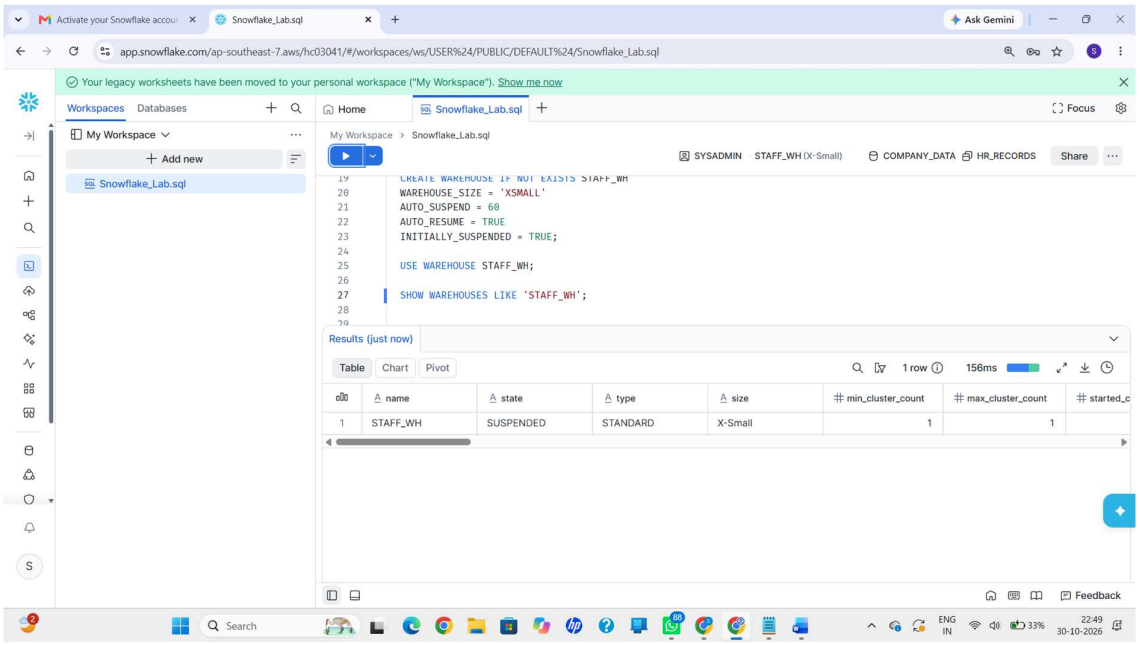
Select the warehouse:

```
USE WAREHOUSE STAFF_WH;
```



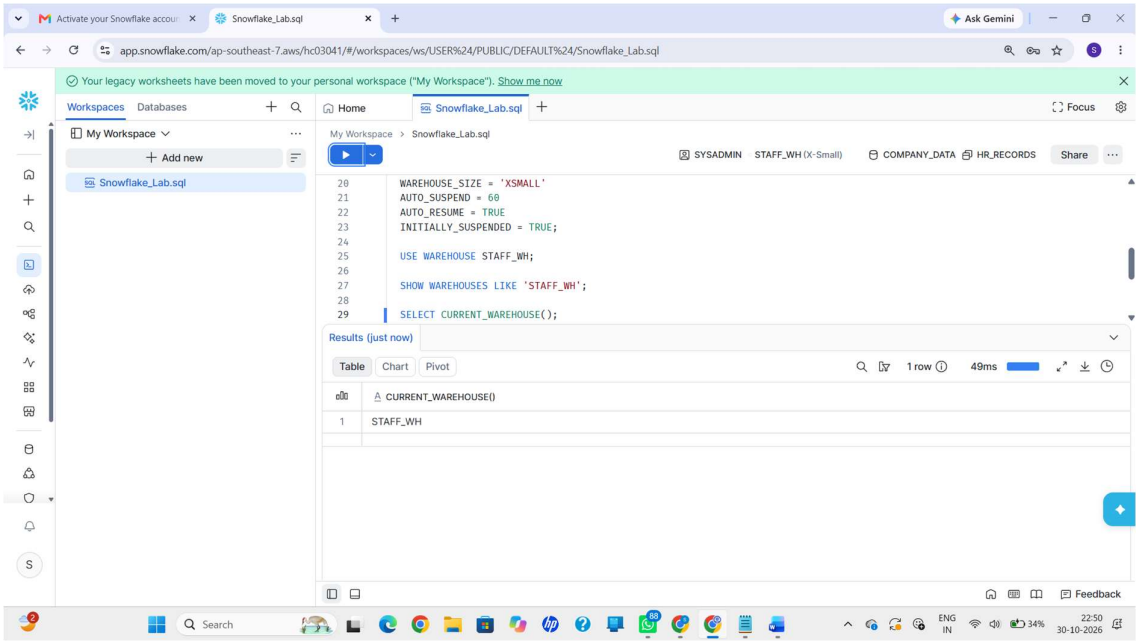
Verify:

```
SHOW WAREHOUSES LIKE 'STAFF_WH';
```



Check active warehouse:

```
SELECT CURRENT_WAREHOUSE();
```



PART 3: TABLE CREATION

Step 5: Create an Employee Table

```
CREATE TABLE IF NOT EXISTS
COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS

(

    EMPLOYEE_ID NUMBER(10,0),

    EMPLOYEE_NAME VARCHAR(100),

    DEPARTMENT VARCHAR(50),

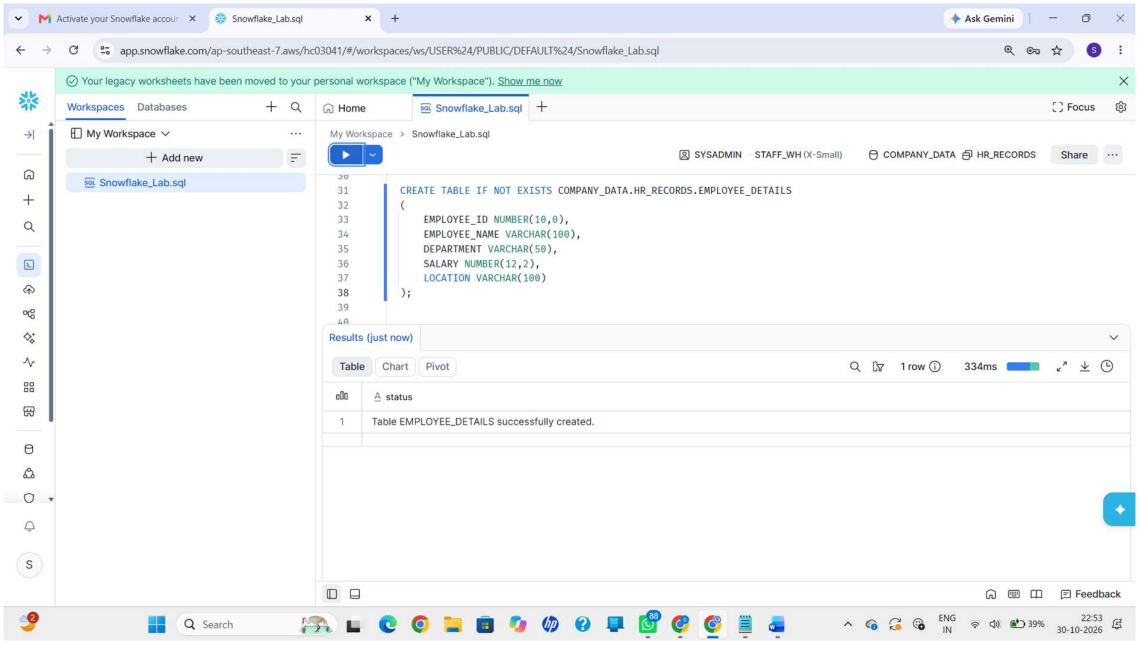
    SALARY NUMBER(12,2),

    LOCATION VARCHAR(100)

);
```

Explanation:

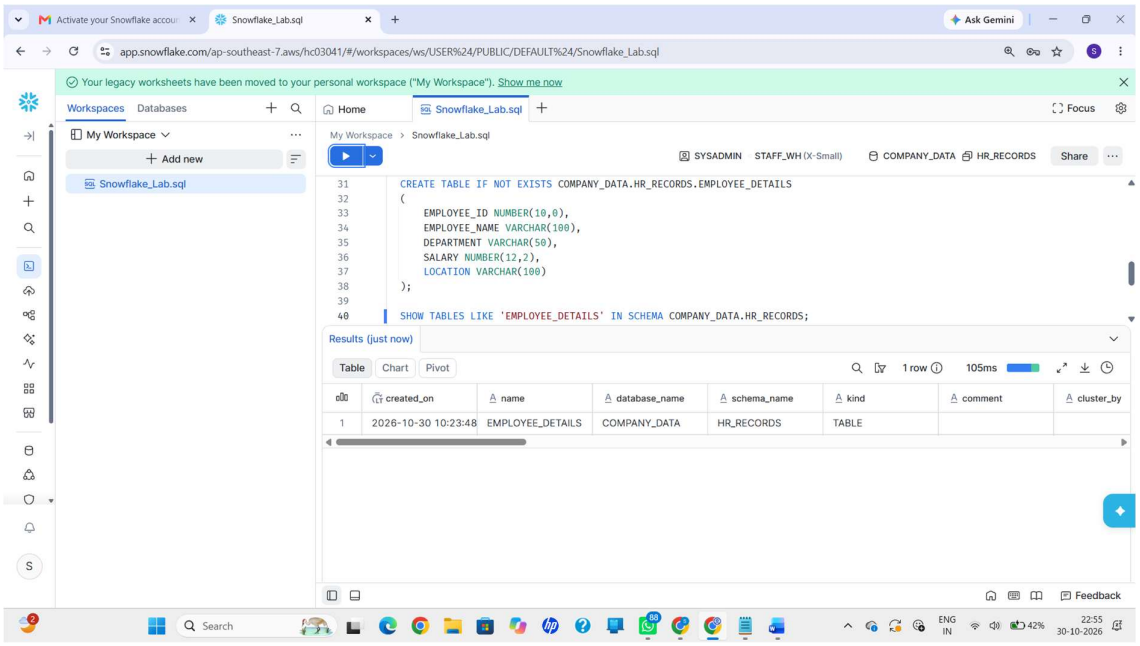
Column	Data Type	Purpose
EMPLOYEE_ID	NUMBER(10,0)	Employee identifier
EMPLOYEE_NAME	VARCHAR(100)	Employee name
DEPARTMENT	VARCHAR(50)	Department
SALARY	NUMBER(12,2)	Employee salary
LOCATION	VARCHAR(100)	Employee location



Verify table:

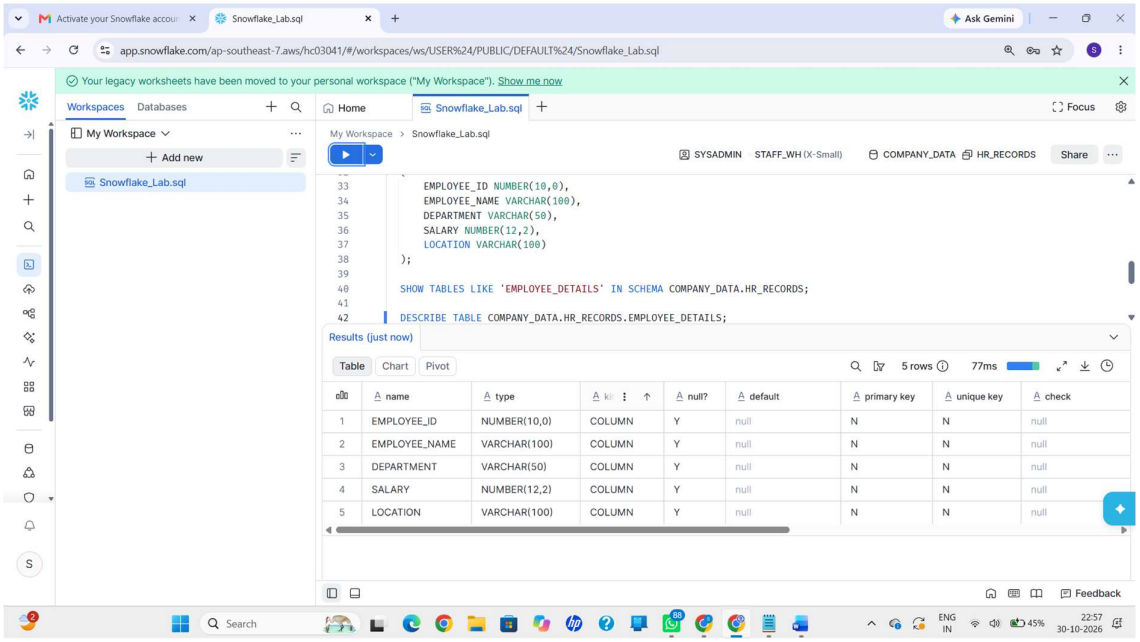
```
SHOW TABLES LIKE 'EMPLOYEE_DETAILS'

IN SCHEMA COMPANY_DATA.HR_RECORDS;
```



Check table structure:

```
DESCRIBE TABLE COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```



PART 4: FILE FORMAT CREATION

Step 6: Create a CSV File Format

A file format defines how Snowflake should read and interpret an external or staged file.

```
CREATE FILE FORMAT IF NOT EXISTS

COMPANY_DATA.HR_RECORDS.STAFF_CSV_FORMAT

TYPE = CSV

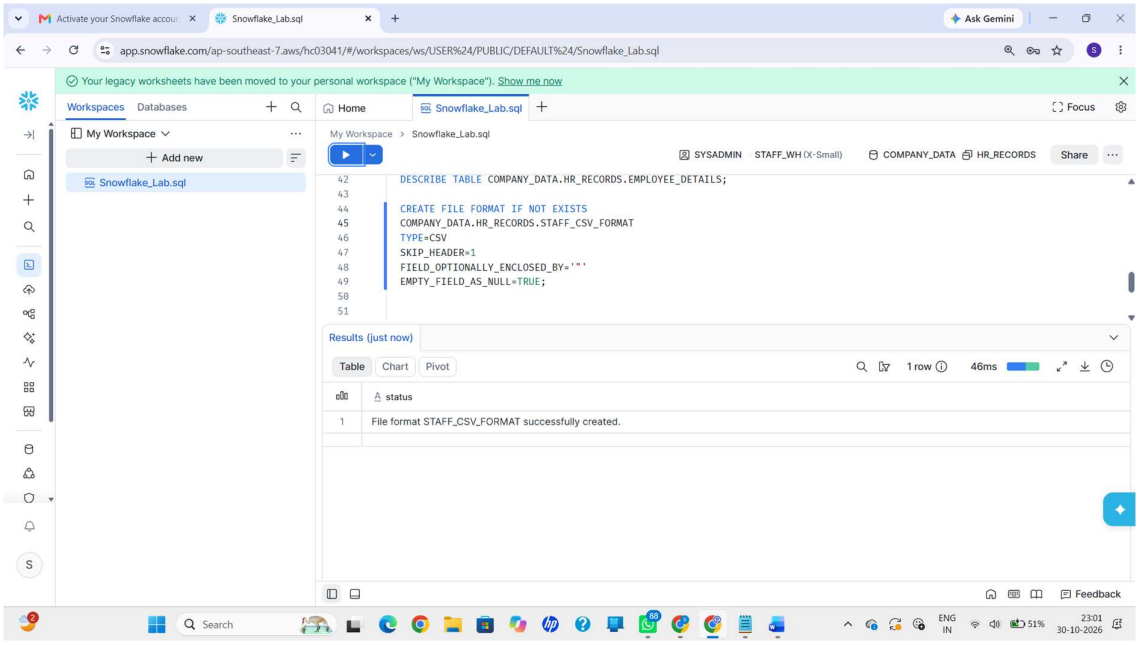
SKIP_HEADER = 1

FIELD_OPTIONALLY_ENCLOSED_BY = '"'

EMPTY_FIELD_AS_NULL = TRUE;
```

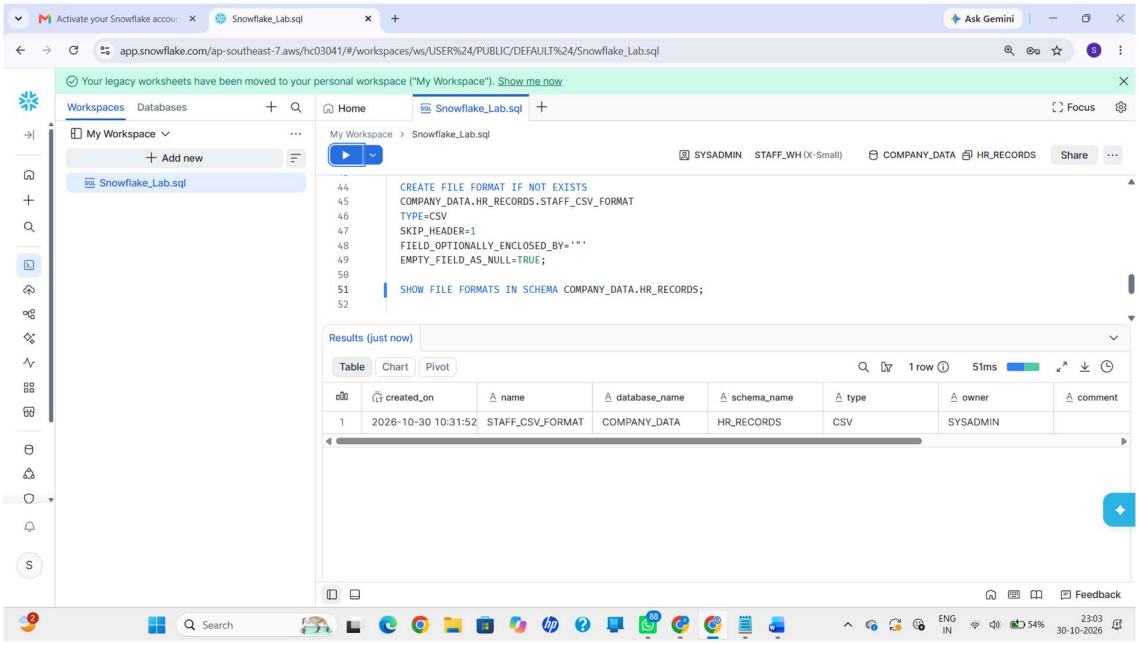
Explanation:

- TYPE = CSV – Specifies CSV format.
- SKIP_HEADER = 1 – Skips the first row containing column names.
- FIELD_OPTIONALLY_ENCLOSED_BY – Handles values enclosed in double quotes.
- EMPTY_FIELD_AS_NULL – Converts empty fields into NULL values.



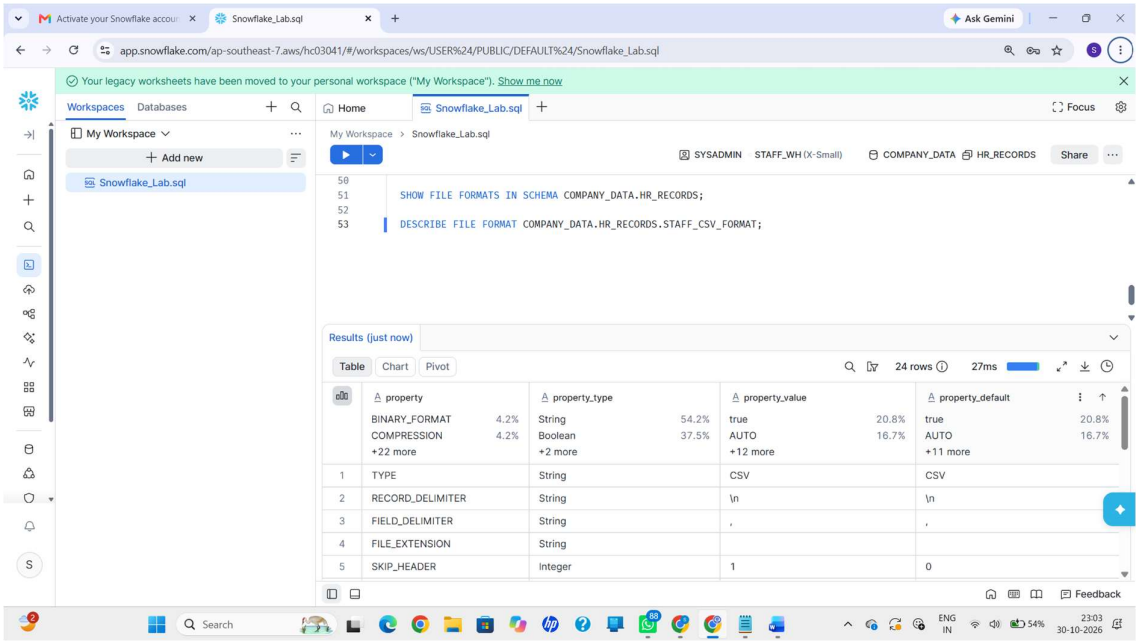
Verify:

```
SHOW FILE FORMATS IN SCHEMA COMPANY_DATA.HR_RECORDS;
```



View file format properties:

```
DESCRIBE FILE FORMAT COMPANY_DATA.HR_RECORDS.STAFF_CSV_FORMAT;
```



PART 5: INTERNAL STAGE MANAGEMENT

Step 7: Create an Internal Stage

A stage is a location used to store or reference files before loading them into Snowflake tables.

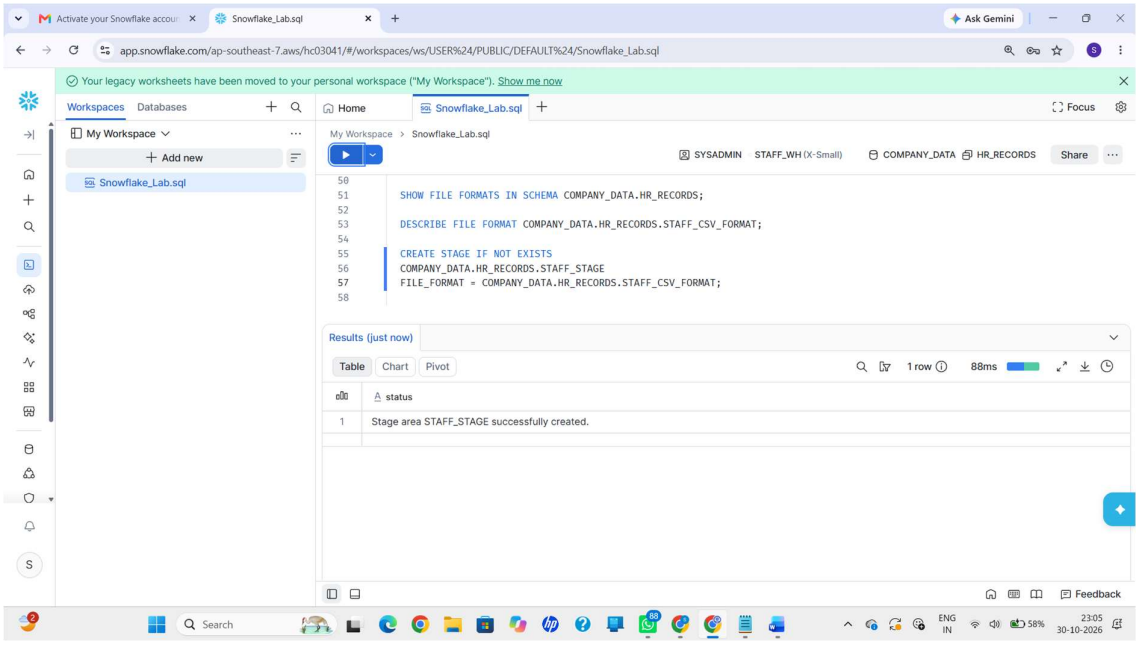
```
CREATE STAGE IF NOT EXISTS

COMPANY_DATA.HR_RECORDS.STAFF_STAGE

FILE_FORMAT = COMPANY_DATA.HR_RECORDS.STAFF_CSV_FORMAT;
```

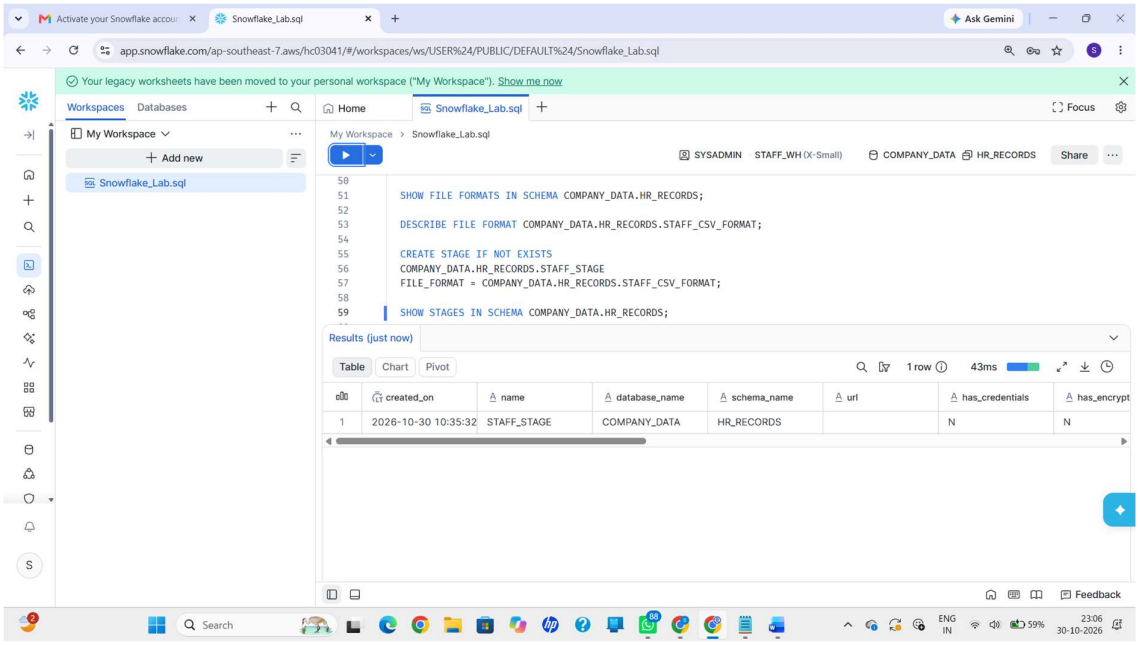
Explanation:

- CREATE STAGE creates a named stage.
- STAFF_STAGE is the stage name.
- FILE_FORMAT specifies the CSV interpretation rules.



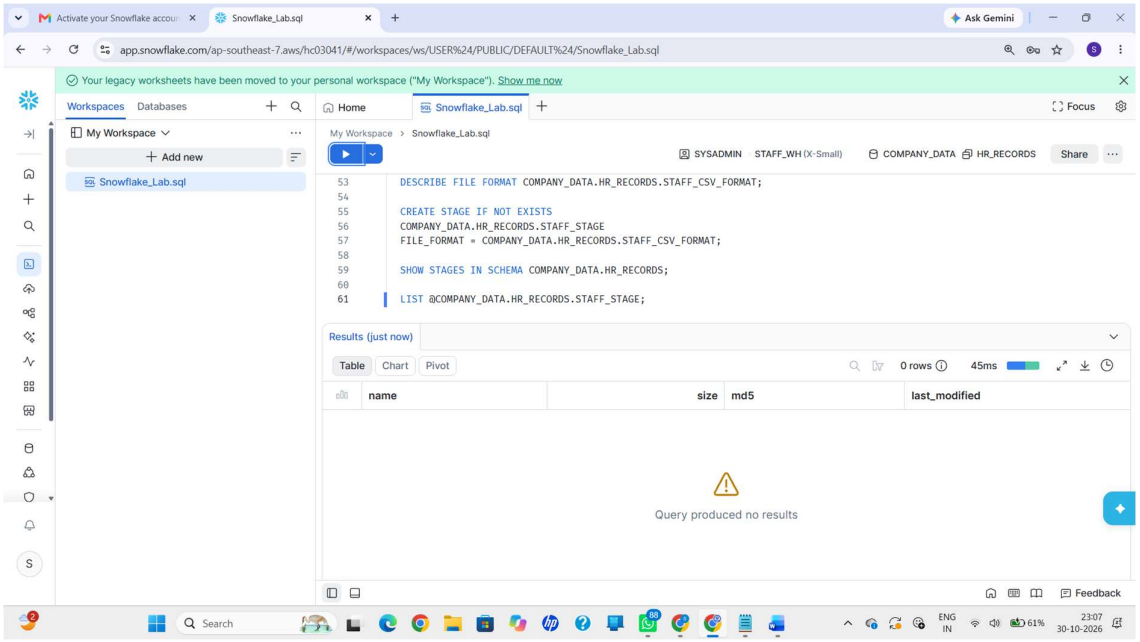
Verify:

```
SHOW STAGES IN SCHEMA COMPANY_DATA.HR_RECORDS;
```



List the files inside the stage:

```
LIST @COMPANY_DATA.HR_RECORDS.STAFF_STAGE;
```



Step 8: Prepare the CSV File

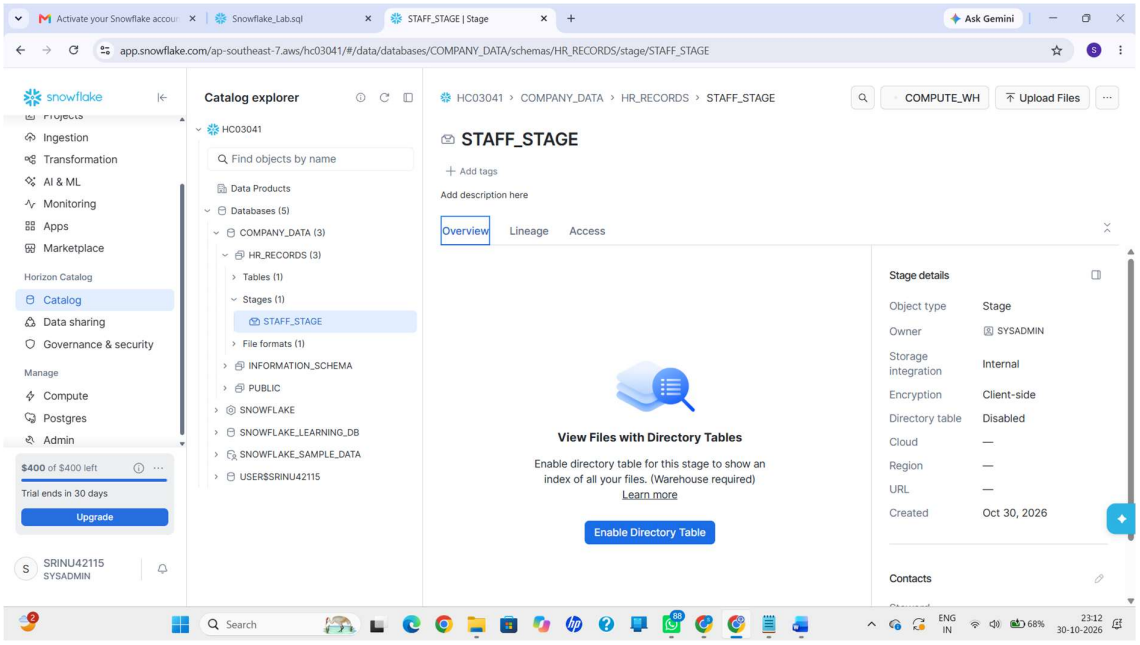
Create a local CSV file named:

staff_records.csv

Add the following data:

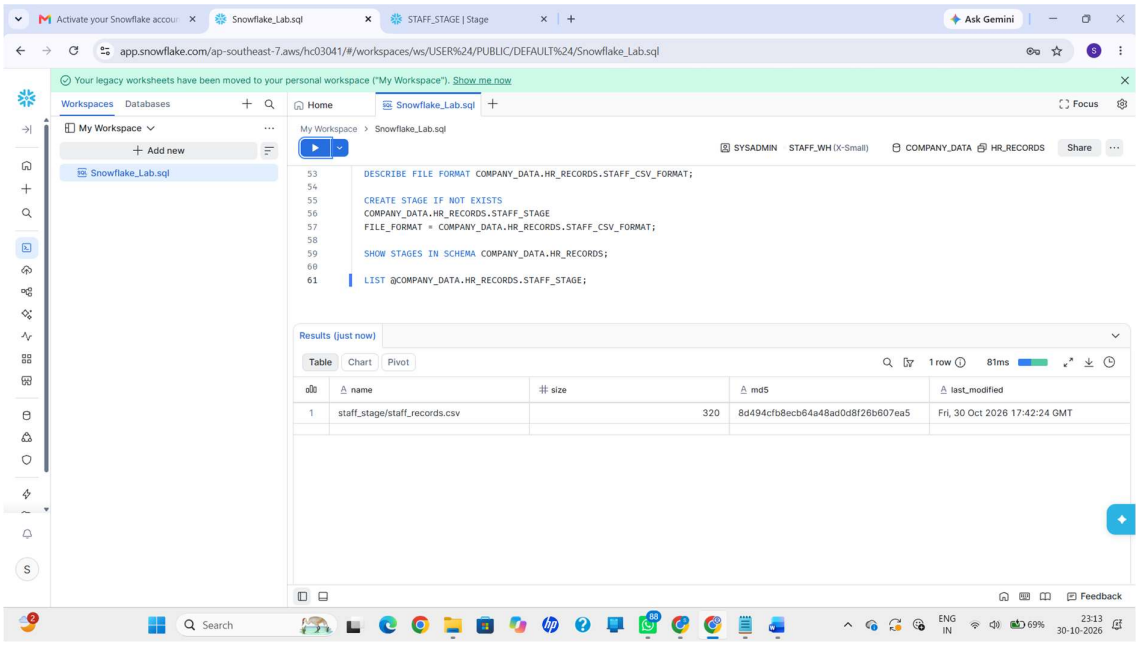
```
EMPLOYEE_ID,EMPLOYEE_NAME,DEPARTMENT,SALARY,LOCATION
2001,Anil,IT,58000,Chennai
2002,Meena,HR,46000,Bangalore
2003,Rajesh,Finance,62000,Hyderabad
2004,Divya,IT,67000,Chennai
2005,Karthik,Sales,42000,Mumbai
2006,Priya,HR,49000,Hyderabad
2007,Suresh,Finance,71000,Bangalore
2008,Nisha,IT,63000,Mumbai
```

Upload staff_records.csv into STAFF_STAGE using the file upload interface in Snowflake Snowsight.



Verify the uploaded file:

```
LIST @COMPANY_DATA.HR_RECORDS.STAFF_STAGE;
```



PART 6: LOADING DATA INTO SNOWFLAKE

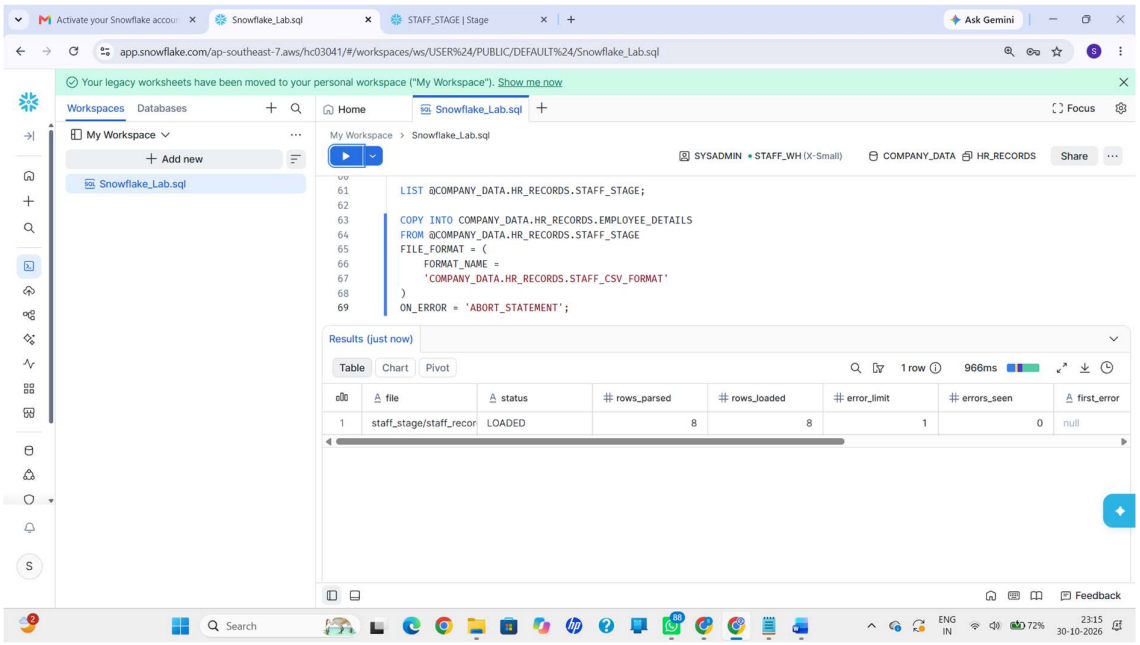
Step 9: Load CSV Data Using COPY INTO

```
COPY INTO COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS
FROM @COMPANY_DATA.HR_RECORDS.STAFF_STAGE
FILE_FORMAT = (
```

```
    FORMAT_NAME =
    'COMPANY_DATA.HR_RECORDS.STAFF_CSV_FORMAT'
)
ON_ERROR = 'ABORT_STATEMENT';
```

Explanation:

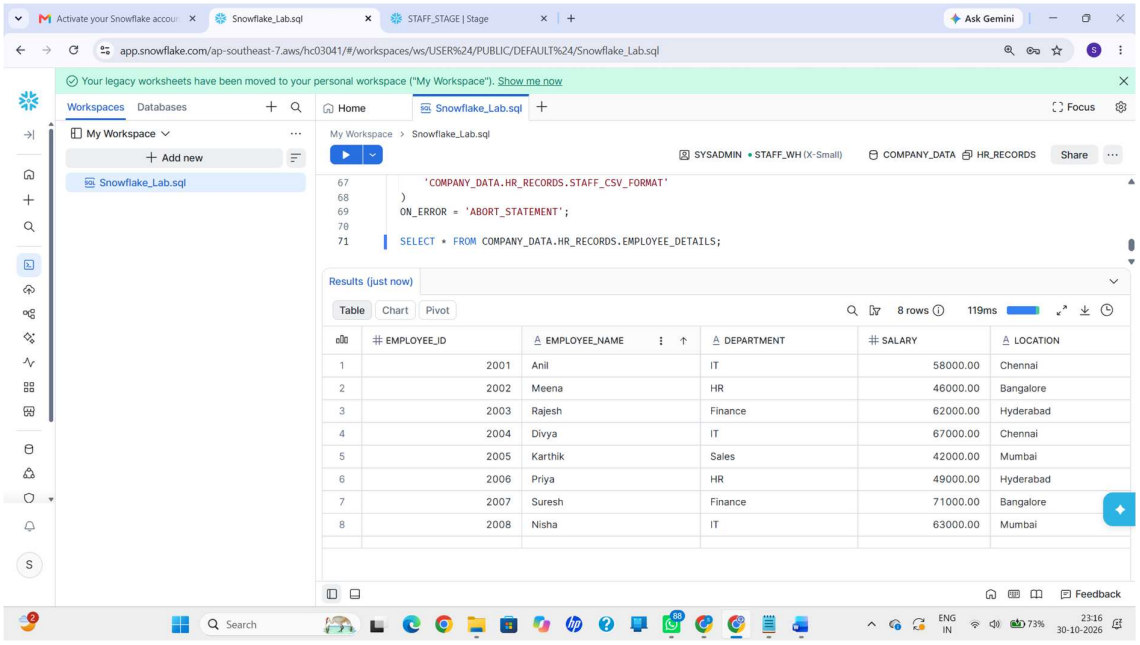
- **COPY INTO – Loads data into a Snowflake table.**
- **EMPLOYEE_DETAILS – Destination table.**
- **FROM – Specifies the source.**
- **STAFF_STAGE – Source stage.**
- **FILE_FORMAT – Specifies the CSV parsing configuration.**
- **ABORT_STATEMENT – Stops the load if an error occurs.**



Step 10: Verify Loaded Records

Display all records:

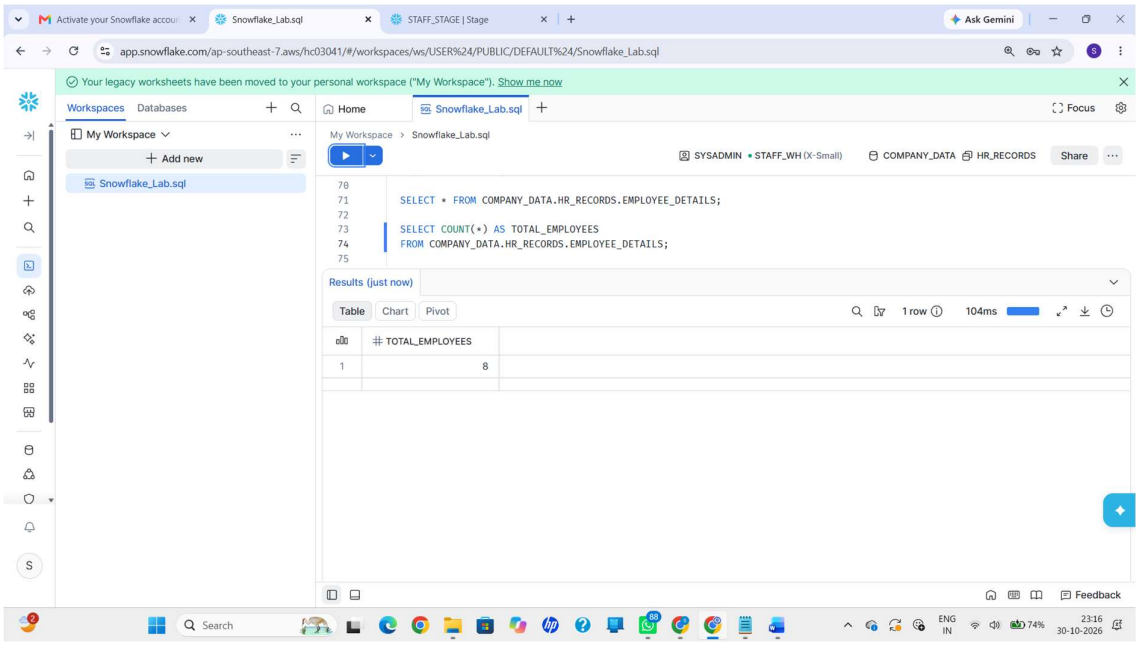
```
SELECT * FROM COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```



Count the records:

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES

FROM COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
```

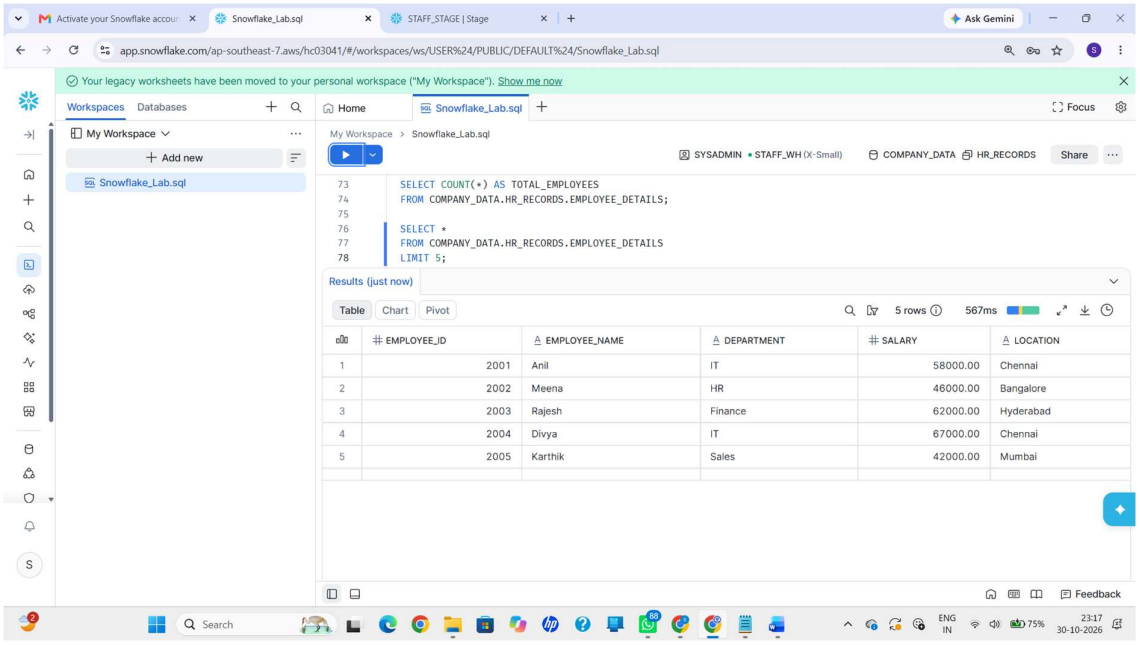


Display the first five records:

```
SELECT *

FROM COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS

LIMIT 5;
```


Activate your Snowflake accour
Snowflake_Lab.sql
STAFF_STAGE | Stage
Ask Gemini
app.snowflake.com/ap-southeast-7.aws/hc03041/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/Snowflake_Lab.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
Snowflake_Lab.sql
Home
Snowflake_Lab.sql
Focus
My Workspace > Snowflake_Lab.sql
SYSADMIN • STAFF_WH (X-Small)
COMPANY_DATA HR_RECORDS
Share
73 SELECT COUNT(*) AS TOTAL_EMPLOYEES
74 FROM COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS;
75
76 SELECT *
77 FROM COMPANY_DATA.HR_RECORDS.EMPLOYEE_DETAILS
78 LIMIT 5;
Results (just now)
Table
Chart
Pivot
5 rows
567ms
EMPLOYEE_ID
EMPLOYEE_NAME
DEPARTMENT
SALARY
LOCATION
1 2001 Anil IT 58000.00 Chennai
2 2002 Meena HR 46000.00 Bangalore
3 2003 Rajesh Finance 62000.00 Hyderabad
4 2004 Divya IT 67000.00 Chennai
5 2005 Karthik Sales 42000.00 Mumbai
Feedback
Search
ENG
IN
75%
23:17
30-10-2026